On the Problem of the In-and-Circumscribed Triangle.

[Continued from p. 216. Vol. VIII.]

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The following is a continuation of the table of cases for the in-and-circumscribed triangle problem. Each row exhibits the specification of the case, the straight sum, the deficiency, and the number of examples.

Table (continued). Pages 217–221.

No. of Case	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
30	$a : b : c,\ a = b = c,\ \alpha + \beta = \gamma$	(15)	(20)	$2(a-1)YZ \cdot Z\alpha\beta$
31	$a : b : c,\ a = b,\ \alpha = \beta = \gamma,$ $\alpha + \beta = \gamma$	6	4	$2x(a-1)YZ \cdot Z\alpha\beta$
32	$a : b : c,\ a = b,\ \alpha = \beta = \gamma,$ $\alpha + \beta = \gamma$	6	4	$2x(a-1)YZ \cdot Z\alpha\beta$
33	$a : b : c,\ \beta = \gamma,\ \alpha + \beta = \gamma$	6	4	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
34	$a : b : c,\ \beta = \gamma,\ \alpha + \beta = \gamma$	6	4	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
35	$a : b : c,\ \beta = \gamma,\ \alpha + \beta = \gamma$	6	4	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
36	$a : b : c,\ \alpha + \beta = \gamma$	(12) (16)		Quartic case

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
37	$a : b : c,\ a = b = c,\ \alpha = \beta = \gamma$	(12)	(16)	$2x(a-1)(X-2) \cdot YZ$
38	$a : b : c,\ a = b,\ \alpha = \beta = \gamma$	6	4	$2X(a-1)(Y-Z) \cdot YZ + Y \cdot V$
39	$a : b : c,\ a = b,\ \alpha = \beta = \gamma$	6	4	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
40	$a : b : c,\ a = b,\ \beta = \gamma$	6	4	$2(a-1)(aX - aZ)YZ \cdot Z\alpha\beta$
41	$a : b : c,\ \beta = \gamma$	3	6	$2X(a-1)(X-2) \cdot YZ$
42	$a : b : c,\ \beta = \gamma$	3	6	$2(a-1)\{aX - 2(X-1)Y(X-Z)\} \cdot YZ$

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
38	$X = D = P = u,\ V = v,\ u = v,$ $P = B = u,\ B = v,\ \beta = u - v,$ $D = P = u,\ D = v$		6	$2X(YpaX + aY)Z \cdot YZ$
39	$X = D = P = u,\ a = b = c,\ \beta = v$	6	3	$\{2X^2 + aZY \cdot 8\beta b(b-c) \cdot a(\beta b)^2 - (aP) \cdot a(\beta b)^2\} \cdot ab$
40	$a : b : c,\ X = P = u,$ $a : b : c,\ V = u$		3	$\{a(\beta x)^2 + (aP)^2 \cdot (\beta b)Z \cdot (\beta b) \cdot Z \cdot aX$
41	$a : b : c,\ \beta = \gamma,\ a = b,\ \beta b = \nu,$ $D = u,\ P = v,\ P = R$		3	$2x(a-1)YZ \cdot Z\alpha\beta$
42	$a : b : c,\ a = b,\ \beta = \gamma,$ $D = u,\ P = v,\ P = R$		6	$2(a-1) \cdot Z \cdot Z \cdot abP \cdot ab\alpha\beta(a-1) \cdot \{aZ + (aP)\} \cdot Z \cdot abP$
43	$a : b : c,\ a = b,\ P = R,\ \beta = \gamma$		3	$2(a-1)YZ(Y-Z) \cdot abP$
		(4) (12)		Quartic case

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No.	*Specification*	*Strt. sum*	*Def.*	*No. of examples*
43	$a:b:c$, $a=b$, $\beta=\gamma$, $P=Q$, $\alpha=\beta$, $D=P$	(12)	(17)	$2(a-1)(aX-aZ)YZ\cdot Z\alpha\beta$
44	$a:b:c$, $\beta=\gamma$, $D=P$	6	8	$2(a-1)\cdot YZ\cdot Z\alpha\beta-aX\beta\cdot aZ\beta b+8XZ\cdot aP$
45	$a:b:c$, $a=b$, $\beta=\gamma$, $\alpha+\beta=\gamma$, $P=Q$	6	5	$2X(a-1)YZ(Y-Z)\cdot abP\cdot YZ$
46	$a:b:c$, $a=b$, $\beta=\gamma$, $\beta=\gamma$	6	5	$2X(a-1)YZ(Y-Z)\cdot abP\cdot YZ$
47	$a:b:c$, $a=b$, $\beta=\gamma$, $\alpha=\beta=\gamma$	6	5	$2(a-1)Y(Y-Z)(Z-aZ)(YZ-Y\beta+aP)\beta$
48	$a:b:c$, $\beta=\gamma$, $\alpha=\beta$	6	5	$2(a-1)Y(Y-Z)(Z-aZ)(YZ-Y\beta+aP)\beta$
		(12) (16)		at infinity

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No.	*Specification*	*Def.*	*No. of examples*
51	F, a, b, $c=a-b=c$		
52	$a=u=b=P\cdot YZ$		$2aY\cdot ab\beta$
			$2X^2(3a^2-9aZ-aP+(Z-aZ))$ $+aX^3(2aZ-aP)(2P-9P-aZ\beta)$ $+aX^3(a^2P-aP\beta-aP\beta)$ $+aX^3(2P-aP\beta)(P-aP)(P-aZ)$ $+aX^3(2P+aP\beta)(P-aZ)$ $+\left(\sum aX^3\right)(P-aZ\beta)$
D ditto		503	

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The foregoing results are chiefly obtained by means of the theory of correspondence; viz. if instead of the triangle $aBcDeF$ we consider the unclosed trilateral $aBcDeFg$, where the points a and g are situate on one and the same curve, say the curve $a = g$, then the points a and g have a certain correspondence, say a (χ, χ') correspondence with each other; and when a, g are a "united point" of the correspondence, the trilateral in question becomes an in-and-circumscribed triangle $aBcDeF$; that is, the number of triangles is equal to that of the united points of the correspondence, subject however (in many of the cases) to a reduction on account of special solutions. It is in particular cases more convenient to consider the correspondence not of the united points in, in several of the cases, but not in all of them, $a = g = \chi = \chi'$. But in some instances I employ a functional method, by assuming that the identical curves are each of them the aggregate of the two curves a, a': we have obtain for the number $g\sigma$ of the triangles belonging to the curve a a functional equation $\phi(a + a') = \phi a + \phi a'$ given function; viz. the expression on the right-hand side depends on the solution of the preceding cases, wherein the number of identities between the several curves is less than in the case under consideration; and taking it to be known, the functional equation gives $\phi a =$ particular solution + linear function of (a, B, F). The particular solution is always easily obtainable, and the constant of the linear function can be determined by means of particular forms of the curve a.

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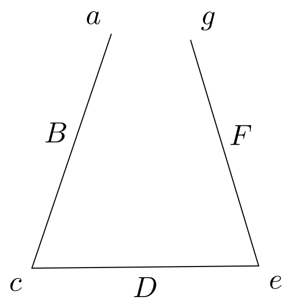
Article Nos. 1 to 6. *The Principles of Correspondence as applied to the present Problem.*

1. Consider the unclosed trilateral $aBcDeFg$, where the points a and g are on one and the same curve, say the curve $a = g$. Starting from an arbitrary point a on the curve, we have αB any one of the tangents from a to the curve B, touching this curve, say at the point B; and intersecting the curve c in a point c; viz. c is any one of the intersections of aBc with the curve c; we have then similarly cDe any one of

Fig. 1.



the tangents from c to the curve D, touching it, say at D, and intersecting the curve e in a point e; viz. the point e is any one of the intersections in question; and then in like manner we have eFg any one of the tangents from e to the curve F, touching it, say at F, and intersecting the curve $\alpha(= a)$ in a point g; viz. g is any one of the intersections in question. Suppose that to a given position of a there correspond χ positions of g, it is easy to find the value of χ; viz. if (as above tacitly supposed)

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the curves a, B, c, D, e, F are all distinct curves, then the number of these tangents αB is $= B$; there are on each of them c points c; through each of these we have D tangents cDe; on each of these e points e; through each of these F tangents eFg; and on each of these a points g; through each of these $\chi = Bcdef\alpha$. But if some of the curves become one and the same curve — if, for instance, $\alpha = B = c$, — the line αBc is here a tangent from a point α on the curve, we exclude the tangent at the point α, and the number of the remaining tangents is $= (A-2)$; each tangent meets the curve in the point α counting once, the point B counting twice, and in $(\alpha - 3)$ other points; that is, the number of the points c is $= (A-2)(\alpha-3)$, and in like cases; the calculation is always immediate, and the only difference is that, instead of a factor α or A, we have such factor in its original form or diminished by 1, 2, or 3, as the case may be. Similarly starting from g, considered as a given point on the curve $g(= a)$, we find χ' the number of the corresponding points a; thus in the case where the curves are all distinct curves, we have $\chi' = FeDcBa\,(= \chi)$; and so in other cases we find the value of χ'. The points (a, g) have thus a (χ, χ') correspondence, where the values of χ, χ' are found as above.

2. There will be occasion to consider the case where in the triangle $aBcDeF$ (or say the triangle $aBcDeFa$) the point a is not subjected to any condition whatever, but is a free point. There is in this case a "locus of a," which is at once constructed as follows: viz. starting with an arbitrary tangent αB of the curve B, touching it at B and intersecting the curve c in a point c; through c we draw the curve D the tangent cDe, touching it at D and intersecting the curve e in a point e; and finally from e to the curve F the tangent eFa, touching it at F and intersecting the original tangent αB in a point α; the locus of the point α so obtained is an arbitrary tangent of the curve B (or of the curve F).

3. The general form of the equation of correspondence is

$$p(a - x - x') + q(b - \beta - \beta') + \ldots = k\Delta(t);$$

viz. if on a curve for which twice the deficiency is $= \Delta$ we have a point P corresponding to certain other points $P', Q', \ldots$, in such wise that P, P' have an (x, x') correspondence, P, Q' a (β, β') correspondence, &c.; and if (α) be the number of the united points (P, P'), (β) the number of the united points (P, Q'), &c.; and if moreover for a given position of P the points $P', Q', \ldots$ are obtained as the intersections of the curve with a curve Θ (depending on the point P) which meets the curve k times at P; p times at each of the points P', q times at each of the

* To avoid confusion with the notation of the present memoir, I abstain in the text from the use of D as denoting the deficiency, and there is a convenience in the use of a single symbol for twice the deficiency; but writing for the moment D to denote the deficiency, I remark, in passing, that perhaps the true theoretical form of the equation is

$$k(2D - D - D) + p(a - x - x') + q(b - \beta - \beta') + \ldots = 0,$$

viz. the point P is here considered as having with itself a (D, D) correspondence, the number of the united points therein being $= 0$.

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unilary values $f = \phi - \phi'$ and $v - e - e'$ are obtained by means of a more simple application of the principle of correspondence, as will appear in the sequel ([1]), but for the moment I do not pursue the question.

Article Nos. 7 to 14. ***Locus of a free angle (α).***

7. I consider the case where a is a distinct curve $\alpha =, \nu F$, and where, as was seen, the equation is simply

$$g - \chi - \chi' = 0.$$

I suppose further that a is distinct from all the other curves, or say, *simpliciter*, that a is a distinct curve. The values of χ, χ' will here each of them contain the factor a, say we have $\chi = n\alpha$, $\chi' = n\alpha'$; and therefore the equation gives $g = \alpha(n + \alpha')$. It is obvious that α, α' are the values assumed by χ, χ' respectively in the particular case where the curve α is an arbitrary line ($\alpha = 1$); and $\alpha + \alpha'$ is the number of the united points on this line.

8. Suppose now that in the triangle $aBcDeFa$ the point a is a free point, and have, as above-mentioned, a locus of a, and the united points on the arbitrary line are the intersections of the line with this locus; that is, the locus meets the arbitrary line in $\alpha + \alpha'$ points; or, what is the same thing, the order of the locus is $= \alpha + \alpha'$.

9. I stop for a moment to remark that in the particular case where the curve B is a point ($B = 1$), then in the construction of the locus of α the arbitrary tangent αB is an arbitrary line through B, and the construction gives on this line a position of the point α. But drawing from B a tangent to the curve F, and thus constructing in order the points F, e, D, c, α, the construction shows that B is an α'-tuple point on the locus; and (by what precedes) an arbitrary line through B meets the locus in α other points; that is, in the particular case where the curve B is a point, the order of the locus of α is $= \alpha + \alpha'$, which agrees with the foregoing result.

10. The construction for the locus of α may be presented in the following form: viz. drawing to the curve D a tangent cDe, meeting the curves c, e in the points c, e respectively; then if from any point c we draw to the curve B a tangent cBa, and from any point e to the curve F a tangent eFa, the tangents cBa, eFa intersect in a point on the required locus. Hence if in any particular case (that is for any particular position of the tangent cDe) the lines cBa, eFa become one and the same line, the point a will be an indeterminate point on this line; that is, the line in question will be part of the locus of a.

11. The case cannot in general arise so long as the curves B, F are distinct from each other; but when these are one and the same curve $B = F$, then the line cBa will be a tangent of the curve B, and we have thus as part of the locus the locus $c = e$ if it can be drawn, the same being supposed to. To show how this is, suppose in, say at F, and intersecting the curve cBe, it is needful, that perhaps the true theoretical form of the equation is[1]

[1] See post, Nos. 24 *et seq.*

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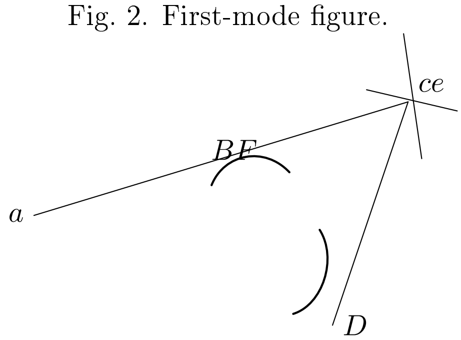
here a tangent at B passing through a point α of the intersection of this curve α, and from this point a tangent drawn to the curve $B = F$. For the position in question of the tangent of D, the points c, e coincide with each other, and we have thus the coincident tangents cBa and eFa as the identical curves $B = F$. It is further

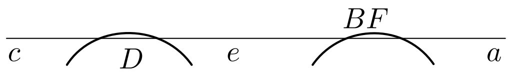
Fig. 2. First-mode figure.



to be remarked that the number of the points of intersection is $= \sigma$; from each of these there are B tangents to the curve F (in all σB tangents), and each of these counts once in respect of each of the D tangents to the curve D; that is, it counts D times. We have thus as part of the locus of α on B lines each D times, or, say, first-mode reduction $= \sigma BD$.

12. The second mode is there in the same manner "second-mode figure." The tangent from D is here a common tangent of the curves D and $B = F$. This meets the curve c in σ points; and the curve e in σ points; and attending to one pair of points c, e on each of these the tangents cBa, eFa, coinciding with the common tangent

Fig. 3. Second-mode figure.



in question, and forming part of the locus of α. The number of the common tangents is $= BD$; but each of these counts once in respect of each combination of the points c, e; that is in all σ times. And we have thus as part of the locus BD lines each σ times, or, say, second-mode reduction $= BD \cdot \sigma$. This is (as it happens) the same number as for the first mode; but to distinguish the different origins I have written as above $\sigma \cdot BD$ and $BD \cdot \sigma$ respectively.

13. It is important to remark that each of the two modes arises whatever relations of identity subsist between the curves σ, c, B, and F, but with consideration of these. Thus if any one of the lines cBa, eFa is not so situate (but is so far) distinct from B, then in the first-mode figure as may be a node or a cusp of the curve $\sigma = \sigma$, it, say at F, and intersecting the curve $\sigma = \sigma$, it is needful to find a copy of the entire of the intersections in question. Suppose that for a given position of D touching the curve $c = e$ in the neighbourhood of the node, thus of the two points of intersection each $c = e$ is in the neighbourhood of the node, thus of the two points of intersection each

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in succession may be taken for the point c, and the other of them will be the point e; so that the node counts twice. It requires more consideration to perceive, but it will be readily accepted that the cusp counts three times. Hence if for the curve $c = e$ the number of nodes be $= \delta$ and that of cusps $= \kappa$, the value of the first-mode reduction is $= (2\delta + 3\kappa)BD$.

As regards the second-mode figure, the only difference is that c, e will be now any pair of intersections (each pair twice) of the tangent with the curve $c = e$; and the value is thus $= (c^2 - c)BD$.

It would be by no means uninteresting to enumerate the different cases, and indeed there might arise some advantage in doing so here; but I have (instead of this) considered the several cases, when and as they arise in connexion with any of the cases of the in-and-circumscribed triangle.

14. Observe that the general result is, that in the case $B = F$ of the identity of the curves B and F, but not otherwise, the locus of α includes as part of itself a system of lines; or, say, that it is made up of the proper locus, the lines, and of a residual curve of the order $\alpha + \alpha' -$ Red., which is the proper locus.

Article Nos. 15 to 17. ***Application of the foregoing Theory as to the locus of*** (α)**.**

15. Reverting now to the case where the angle a is not a free angle but is situate on a given curve α, then if the curve α is distinct from the curves α, F, the number of positions of α is, as was seen, $g = \chi + \chi'$. But the points in question are the intersections of the line-system $\alpha F \gamma$ with the curve α; and the line-system $\alpha F \gamma$ being made up of the proper locus $-$ Red. and of a system of lines is on this account distinct from the curve α; the points of intersection on the proper locus must convenient to obtain directly one than the other of two reciprocal results; for instance, to consider the case $B = D = F$ rather than $\alpha = e = e$, since in the equation in the $\chi = \chi'$ Red., the entry in the expression $\alpha + \alpha' -$ Red., the order of the proper locus of α.

16. It is however to be noticed that if the curve α, being as is assumed distinct from the curves α, and $F = B$, is identical with one or both of the remaining curves c, D, the foregoing expression $\chi + \chi'$ for the order of αBe, gives points a which are not true solutions of the problem, viz. the curve α may include in general in eaiser and more convenient to obtain directly one than the other of two reciprocal results; for instance, of lines, or reduction in the expression $\alpha + \alpha' -$ Red. of the order of the proper locus of α.

[1] More generally, if the curve α is a curve identical with any of the other curves, then if treating in the first instance the angle α as the free we find in any manner the locus of α, the required positions of the angle α are the intersections of this locus and of the curve α; but these intersections will in general include intersections which give heterotypic solutions. The determination of these is a matter of some delicacy, and I have in general treated the problem in each manner that the question does not arise; but an an example see post, Case 45.

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17. But this cannot happen if the curve a is distinct also from the curves c, D; or, say, simply when a is a distinct curve. The conclusion is, that in the case where a is a distinct curve we have

$$g = \chi + \chi' - \text{Red.},$$

where the term "Red." vanishes except in the case of the identity $B = F$ of the curves B, F; and that when this identity subsists it is $= \sigma$ times the reduction in the order of the locus of a considered as a free angle; viz. this consists of a first- mode and a second-mode reduction as above explained.

Article Nos. 18 to 21. *Remarks in regard to the Solutions for the 52 Cases.*

18. Before going further I remark that the principle of correspondence applies to corresponding and united tangents in like manner as to corresponding and united points, and that all the investigations in regard to the in-and-circumscribed triangle might thus be presented in the reciprocal form, where, instead of points and lines, we have lines and points respectively. But there is no occasion to employ any such reciprocal process; the result to which it would lead is the reciprocal of a result given by the original process, and as such it can always be obtained by reciprocation of the original result, without any performance of the reciprocal process.

19. It is hardly necessary to remark that although reciprocal results would, by the employment of the two processes respectively, be obtained in a precisely similar manner, yet that this is not so when only one of the reciprocal processes is made use of; so that, using one process only, it may be in general in easier and more convenient to obtain directly one than the other of two reciprocal results; for instance, to consider the case $B = D = F$ rather than $a = e = e$. Forcing the analogy in such a case, to consider the case $B = D = F$ rather than $a = c = e$, and having obtained the one result, directly to deduce from it the other by reciprocity; but that it may nevertheless be interesting to obtain each of the two results directly.

20. It is moreover obvious that although the several forms of the same case, for instance Case 2, $a = c$, $a = e$, or $c = e$, are absolutely equivalent to each other, yet that, when as above we select a vertex a, and seek for the number of the united points (a, g), the process of obtaining the result will be altogether different according to the different form which we employ. For instance, in the case just referred to, if the form is taken to be $a = c$ or $c = e$, then the equation $g = \chi + \chi'$ is applicable to it; but not so if the form is taken to be $a = e$. It would by no means uninteresting in every case to consider the several forms successively; but it appears that in every case it cannot however be carried out very far, considering, illustration, verification, or otherwise it appears proper so to do. The translation of a result, for instance, of a form $a = c$ or $c = e$ into that for the form $a = c = e$ is so easy and obvious, that it is not even necessary formally to make it.

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21. I do not at present further consider the general theory, but proceed to consider in order the 52 cases, interpreting in regard to the general theory each further discussion or explanation as may appear necessary. In the several instances in which the equation $g = \chi + \chi'$ is applicable, it is sufficient to write down the values of χ, χ', the mode of obtaining these being already explained.

The 52 Cases for the in-and-circumscribed triangle.

Case 1. No identities.

$$\chi = BcDeFa, \quad \chi' = FeDcBa\,(= \chi),$$

$$g = 2aBcDF.$$

Case 2. $a = c = e$.

$$\chi = B(a-1)DeFa, \quad \chi' = FeD(a-1)Ba,$$

$$g = 2a(a-1)BDF.$$

Second process, for form $a = c = e$. The equation of correspondence is here

$$g - \chi - \chi' + F(a - c - c') = 0;$$

but the points e being given as all the intersections of the curve $a(= c)$ by the line- system cDe which does not pass through a, we have $c = c' = 0$, so that $g = \chi + \chi'$; and then

$$g = BcDeFa(= 2a-1) + FeDcBa,$$

$$= 2a(a-1)BDF,$$

giving the former result([1]).

Case 3. $B = F = e$. Reciprocation from 2; or else, second process,

$$\chi = DeBa(B-1)a, \quad \chi' = De(B-1)Ba,$$

$$g = 2X(X-1)Daa.$$

Third process: form $F = Ba = e$. We have here $g = \chi + \chi' -$ Red.,

$$\chi = BcDaBa, \quad \chi' = BcD(\alpha)a,$$

$$\chi + \chi' = 2X^2Dca.$$

and the reductions are those of the first and second mode, as explained ante, Nos. 13, 14, viz. each of them is $= X \cdot Dca$, and together they are $= 2XDca$; whence the foregoing result.

Case 4. $a = B = a$.

$$\chi = BcDeFa, \quad \chi' = FeDc(\alpha-2)a,$$

$$g = 2XaeDF.$$

[1] Of course, the result is obtained in the form belonging to the next form of specification, viz. here it is $= 2(a-1)BDF$, and as in other instances; but it is unnecessary to refer to this change.

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Observe this is what the result for Case 1 becomes on writing therein $a = B = a$, viz. the opposite curves a, D may become one and the same curve without any alteration in the form of the result.

Case 5. $a = B = a$.

$$\chi = (X-2)cDeFa, \quad \chi' = FeDcXa(X-1),$$

where

$$(X-2) + x \cdot X(x-1) = X(x-1) = X(X-x-1);$$

therefore

$$g = X(Xa - X - a)cDF.$$

Case 6. $a = c = e = a = a$: perhaps most easily by reciprocation of Case 5. The triangle $aBcDeF$ is here in succession each of the eight triangles

$$\begin{array}{cccc} a\ B\alpha\ D\alpha\ F & a\ B\alpha'\ D\alpha\ F & a\ B\alpha'\ D\alpha\ F & a\ B\alpha\ D\alpha'\ F \\ a'\ B\alpha\ D\alpha'\ F, & a\ B\alpha'\ D\alpha\ F, & a\ B\alpha'\ D\alpha\ F, & a\ B\alpha\ D\alpha'\ F, \\ a\ B\alpha\ D\alpha\ F, & a'\ B\alpha\ D\alpha'\ F, & a'\ B\alpha\ D\alpha\ F, & a'\ B\alpha\ D\alpha\ F, \\ a'\ B\alpha\ D\alpha'\ F, & a'\ B\alpha'\ D\alpha\ F, & a'\ B\alpha'\ D\alpha\ F, & a\ B\alpha'\ D\alpha\ F, \end{array}$$

where the two top triangles give Δa and $\Delta a'$ respectively; the remaining triangles all belong to Case 2, and those of the first column give each $2(a^2 - x)BDF$, and those of the second column each $2(a^2 - x)'BDF$. We have then

$$\phi(a + a') = \Delta a + \Delta a' + \{4(a^2 + aa') - 12a\alpha'\}BDF,$$

which is twice the function in question, on the right-hand side we may write

$$\phi a = 2a^2 - 2x'\alpha = \Delta x + a(BX + a)BDF,$$

that is to say so that, a, B are independent of the curves B, D, F; and taking each of these to be a point, and the curve $a = c = e$ to be a conic, this is known that $\phi = 2$; we have therefore $\Delta = 12 + 8a$ in this case, that is $a + B = 2$.

The case where the curve $a = c = e$ is a line gives $\phi = 2 : 4 + a = 2$, that is $a = 2$, let it is not easy to find another condition; assuming however $\eta = 0$, we have $a = 2$, $B = -1$, and hence therefore

$$\Delta a = 2a(a-1)(2a-1) + a(B-1)BD,$$

or my

$$g = 2(a-1)(a-2) + aD,$$

this is a good easy example of the functional process, the use of which appears to exhibit itself; and I have therefore given it, notwithstanding the difficulty as to the complete determination of the constants.

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Third process. The equation of correspondence is

$$g - \chi - \chi' + F(a - c - c') = 0,$$

but for the correspondence (a, c) we have

$$c - c' = v + D(a - y - y') = 0,$$

and for the correspondence (a, c),

$$c - c' = g' - aDa,$$

whence

$$g = \chi + \chi' + aDF \cdot \Delta;$$

that is

$$g = B(a-1)D(a-1)Fa + F(a-1)D(a-1)Ba + 2(a-1)(a-1)y;$$

Moreover

$$\chi + \chi' = B(a-1)D(a-1)Fa \cdot Z,$$

$$\Delta = X - 2a + 8 + a,$$

(if a be the number of cusps of the curve $a = c = e$), and the resulting value is

$$g = 2\{(a-1)BDF \cdot a - 2\}.$$

[Author's remark on the value of $aBDF$:]

[F as the number of cusps of the curve $a = c = e$, the resulting value of g is

$$g = 2(a-1)BDF \cdot \alpha.$$

Where, however, the term aBDF is to be rejected, I cannot quite explain this; I should rather have expected a rejection $= 2aBDF$, introducing the term $-aBDF$. For consider a tangent from the curve D from a cusp of the curve $a = c = e$; there are D such tangents; each gives in the neighbourhood of the cusp two points, say c, e; and from these we draw B tangents cBa to the curve B, and F tangents eFa to the curve F; we have then in respect of the given tangent of D, BF positions of the α tangent of D, BF positions of a, in all $aBDF$ positions of which b corresponds together with the cusp; and for all the cusps together $aBDF$ such triangles; this would be what is wanted; the difficulty is that as if the two intersections at the cusp each in succession may be taken for c, and the other of them for α, it would seem that the foregoing number $aBDF$ should be multiplied by 2.]

Case 7. $B = D = F = a$. Here $g = \chi + \chi' -$ Red.

$$\chi = Xc(X-1)e(X-2)a, \quad \chi' = Xc(X-1)e(X-2)a,$$

that is

$$\chi + \chi' = 2Xa \cdot ce.$$

The reductions of the two modes are as above, with only the variation that in the present case B is the same curve with the two curves $B = F$. That of the first mode is $= X(X-1) \cdot ce$, and that of the second mode is $= (2e-1) \cdot Xce$, or substituting, we have together they are $= 2X(X-1) \cdot ce$; or substituting, we have

$$g = 2X(X-1)X(X-2) + ce.$$

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Case 8. $a = c = B = a$.

$$\chi = (X-2)(a-1)DeFa, \quad \chi' = FeDa(X-1)(a-1)a,$$

$$g = Xc(a-1)(X-1)DeF.$$

Case 9. $D = F = a = a$. By reciprocation of 8.

$$\chi = X(X-1)(X-2)cDF.$$

Case 10. $a = c = B = a$.

$$\chi = B(a-1)(X-2)DeFa, \quad \chi' = FeX(a-2)Ba(a-1),$$

$$g = X(x-1)X(a-1)DeF.$$

Case 11. $D = F = a = a$. By reciprocation of 10.

$$\chi = X(X-1)(X-2) \cdot caa.$$

Second process: form $a = B = a = a$.

$$\chi = (X-1)(a-1)caXa, \quad \chi' = FeXa(X-1)(a-1) \cdot a;$$

giving the former result.

Case 12. $a = c = e = a$, $B = F$.

$$\chi = BcD(a-1)Fa, \quad \chi' = FaT(a-1)Ba\,(=\chi),$$

$$g = 2(a-1) \cdot BDF.$$

Case 13. $F = B = a$, $a = c = e = a$. By reciprocation of 12.

$$\chi = 2X(X-1) \cdot DF.$$

Case 14. $a = c = e$, $B = F = a$.

$$\chi = X(X-1)(a-1)caFa, \quad \chi' = FaT(a-1)F(a-2)\,(=\chi),$$

$$g = 2X(X-1)Y(F-1) + \cdot ce.$$

Case 15. $F = B = a$, $D = F = a$. By reciprocation of 14.

$$\chi = X(X-1)(X-2)Y \cdot DF.$$

Case 16. $a = c = e$, $B = F = a$.

$$\chi = B(a-1)(X-1)caa, \quad \chi' = F(a-1)(X-2)Ba\,(=\chi),$$

$$g = Xa(a-1)Y(X-1).$$

Case 17. $a = c = e$, $B = F = a$.

$$\chi = B(a-1)(X-1)(B-1)ca, \quad \chi' = Fa(a-1) \cdot caR(a-1)\,(=\chi),$$

$$g = 2X(a-1)Y(X-1)a.$$

But we have here aD as an axis of symmetry, so that triangle is counted twice, or the number of distinct triangles is $= \frac{1}{2}g$.

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Case 18. $a = D = c$, $c = B = g$.

$$\chi = Y(g - X)caFa, \quad \chi' = FaY(Y - X)caaa,$$

$$g = 2aXY(Y - y)cF.$$

Case 19. $c = B = a$, $D = F = e$.

$$\chi = Y(g - 1)D(Y - X)c = ea, \quad \chi' = Xy(Y - X)c = Da,$$

$$g = 2aYYY.$$

Case 20. $c = B = a$, $D = F = e$.

$$\chi = Bc(Y - X)cYY = ea, \quad \chi' = Y(Y - X)YDYca,$$

$$g = 2(Y - X)\{aXY + aY + aZ\} - X(Y - X)Yca + 2aZF.$$

Case 21. $c = B = a$, $D = F = e$.

$$\chi = Yc(Y - X)caXa, \quad \chi' = XyY(Y - X)caYa,$$

$$g = 2YcaYY.$$

Case 22. $a = D = c = a = e$, F.

$$\chi = c(X - 1)Da = Fa, \quad \chi' = Fc(X - 1)cDya,$$

$$g = 2Y(X - 1)c \cdot c = F.$$

Case 23. $a = D = c = a = e$, F.

$$\chi = (Y - 2)c(Y - 1)D(Y - 1)y \cdot Fa, \quad \chi' = Fa(Y - 2)cD(Y - 1)Y \cdot ca,$$

$$g = 2(Y - X)c(Y - X - 1)(Y - 2)\{(Y - X)cF\} + 2(Y - 1)yF + aXY + aYZ\} \cdot ZF \cdot \tan;$$

this is then the symmetry of the result of two modes; or any other case.

Case 24. $a = D = c$, $D = c = e = Fa$.

$$\chi = Yy(X - 1)Da = Fa, \quad \chi' = Yy = X(X - 1) \cdot ea\,(= \chi),$$

$$g = 2c(X - 1)Y(Y - 1)y \cdot ca.$$

Case 25. $c = e = a = a = D = F = v$. Here

$$g = \chi + \chi' - \text{Red.},$$

$$\chi = Xy(X - 1)(Y - 1)ca, \quad \chi' = XyY(X - 1)\{(Y - 1)(Y - 1)\}(= \chi)\, e \cdot a;$$

$$\chi + \chi' = v(y - 1) \cdot 2XY(X - 1) \cdot ca.$$

The reductions are those of the first and second modes; $c = e$, and that the curve D is identical with the curves $B = F$.

First-mode reduction is

$$v(C + 2g + 5\kappa)B(B - 1)$$

(where δ, κ refer to the curves $c = e$), which is

$$= v(c - 1)BB(B - 1);$$

that is, the reduction is $= v(g-1)Y(X-1)$.

Second-mode reduction is

$$v(2c+5c)c(X-1)$$

that is, the reduction is $= v(c-1)y(X-1)\cdot c\cdot(Y-1)$;

Hence the two together are $= v(c-1)\{2Y(X-1)+y(Y-1)y\}\cdot$; and subtracting from $\chi+\chi'$ we have

$$g = v(g-1)\cdot\{2X(X-1)-c\}\, c.$$

but on account of the symmetry each triangle is reckoned twice, and the number of triangles is $= \frac{1}{2}g$.

Case 26. $a = c = B = a$, $D = e = F = v$. By reciprocation of 25.

$$\chi = Y(c-1)(Y-1)2 - y(Y-X)Y\cdot cy.$$

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Case 26. $a = c = e = a$, $B = D = v$, $u = F = v$.

$$\chi = Y(a-1)(Y-1)e \cdot 2 - y \cdot Fa, \quad \chi' = Z(Y-1)Y(Y-2)Y(Y-1)\,(\,a-1)\,(\,a-2);$$

$$\begin{aligned} g &= v(a-1)Y(Y-1) \cdot e \cdot F\,(Y-1) \\ &= 2c(a-1)Y(X-1)(a-2) + F(2-A); \end{aligned}$$

Case 27. $c = e = a$, $B = F = a$, $a = D = v$. By reciprocation of 26.

$$\chi = Y(a-1)Y(Y-1) - y(Y-X)Y \cdot xY = 2v\,(\,a-1)\,(\,Y-X)yF.$$

where each triangle counted is twice, so that the number is really one half of this.

Case 28. $a = c = B = a$, $D = e = F = v$.

$$\chi = Yc(Y-1)(Y-2)e \cdot Z, \quad \chi' = Y(Y-1)y \cdot Yc(Y-2)e = v;$$

The reductions are those of the first and second modes; as explained above, with the variation that the curves c and a are here identical, $a = c$, and that the curve D is identical with the curves $B = F$.

First-mode reduction is

$$v(C + 2\delta + 3\kappa)B(B-1)$$

(where δ, κ refer to the curve $c = a$), which is

$$= v(c-1) \cdot Y(Y-1);$$

that is, the reduction is $= v(g-1) \cdot Y(Y-1)$.

Second-mode reduction is

$$v(c-2c)c(X-1)$$

that is, the reduction is $= v(g-1) \cdot Y(Y-1)$.

Hence the two together are $= v(g-1)\{2X(X-1) - c\} \cdot e$, and subtracting from $\chi + \chi'$ the value of g is

$$g = v(g-1) \cdot \{2Y(Y-1) - e\} \cdot e.$$

but on account of the symmetry we must divide by $\frac{1}{2}g$.

Case 29. $c = a = D = v$, $a = c = v$. By reciprocation of 28.

$$\chi = 2X(X-1) \cdot v(v-1) \cdot a.$$

Case 30. $a = c = B = a$, $D = e = v$, $u = F = v$. By reciprocation of 24.

$$\chi = Z(a-1)(Y-1)(Y-2) \cdot eYF \cdot eYF = v \cdot a\,(\,Y-1)\,Y\,(\,Y-2)Y\,F.$$

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Second process. Taking the form

$$c = B = u = a, \quad D = F = u;$$

here

$$g = \chi + \chi' - \text{Red.},$$

$$\chi = Yc(X-1)Da \cdot Ya, \quad \chi' = Yc(X-1) \cdot c \cdot Ye,$$

and

$$\chi + \chi' = 2Y^2 v(a-1)D \cdot e\,(X-1).$$

There is a first-mode reduction,

$$aY\{2v + 3v \cdot (B-1) \cdot v + aZ(X-Y)$$

$$+(X-3)v(v-c-1)Y \cdot a \cdot \beta\}$$

$$+(X-3)v\,(X-2Y\,)$$

$$+(X-3)\,(Y\,Y\,Y);$$

which is

$$= aY\{X(2v-c-1)\} + v \cdot a \cdot \alpha;$$

and a second-mode reduction is

$$= aYX\,(\,X\,(\,Y\,(\,c-1)\,Y\,(c-1) + 12c) \cdot c$$

Hence the two together are

$$= aY\{X(2v-c-1)\} + v(c-1) \cdot (Y - c\,(\,Y-c-1)\,),$$

whence the result is

$$= 2(P-Y)\,(\,c\,(\,c-1)\,Y\,(\,Y-c\,),$$

which agrees with that obtained above.

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 31. $a = D = F = a$, $u = v = v$. By reciprocation of 30.

$$\chi = 2X(X-1)(X-2) \cdot v(v-1)y.$$

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 32. $a = c = v$, $D = e = u$. By reciprocation of 24.

$$\chi = 2X\,(\,X-1)(X-2)\,v\,(\,v-1)\,c.$$

Case 33. $u = c = v$, $a = D = u$.

$$\chi = Y(Y-1)c\,(\,Y-2)\,ay, \quad \chi' = F\,(\,Y-1)e \cdot D\,(\,Y-c\,)\,ya \cdot ZY + Y\,F.$$

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Case 33. $u = c = v$, $a = D = u$.

$$\chi = Y(Y-1)c\,(\,Y-1)\,(\,a-1)\,a\,Fa, \quad \chi' = F\,(\,Y-1)\,e \cdot D\,(\,Y-2)\,a \cdot (\,X-2);$$

$$g = 2X\,(\,X-1)\,(\,X-2)Y\,c\,DF.$$

Case 34. $u = e = v$, $F = u$.

$$\chi = (X-2)c\,(X-1)De \cdot ya, \quad \chi' = Y e Y\, F\, e\, Dc \cdot Xa\,(X-1);$$

$$g = Y\,(\,c-1)Y\,(\,X-1)\,(\,Y-2)\,Da.$$

Case 35. $u = a = v$, $B = F = u$. By reciprocation of 34.

$$\chi = 2Y\,(\,Y-1)\,(\,Y-2)\,F\,(\,Y-2)\cdot cBe.$$

Case 36. $a = c = v$, $B = F = u$.

$$\chi = B\,(\,X-1)\,(\,X-2)\,(\,X-1)\,a\,Fa, \quad \chi' = F\,(\,X-1)Y\,X\,(\,X-1)\,(\,X-2)\,a;$$

$$g = Y\,(\,c-1)\,(\,X-2)\,Y\,F.$$

Case 37. $a = c = v$, $D = v$, $F = v$. By reciprocation of 36.

$$\chi = Y\,(\,X-1)\,(\,X-2)\,Y\,(\,X-1)\,(\,X-2)\,Y\,DBe + 2YX\,F.$$

Case 38. $B = D = v$, $a = v$, $F = v$.

$$\chi = Y\,(\,X-1)Y\,Fa, \quad \chi' = Y\,(\,X-1)\,e\,F\,(\,X-1)\,a;$$

$$g = 2Y\,(\,X-1)Y\,F.$$

Functional process; the curve c is assumed to be the aggregate of two curves, say $a = c + c'$. Drawing the construction…

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which is belongs; thus $aXcDcF$ is $B = c = v = v$, which is Case 10, and so in the other instances. Observing that cases 10 and 14 occur each twice, we have

$$\phi(c+c') - \phi c - \phi c' = DF \text{ multiplied by}$$

$$k(c-1)(X-X+v)\alpha \ldots\ldots (19) \times 2$$
$$+2c(c-1)(v-X-1) \ldots\ldots (8)$$
$$+4c(c-1)(X-X-1) \ldots\ldots (14) \times 2$$
$$+2c(c-1)F \ldots\ldots (12)$$
$$+2c(c-2)F \ldots\ldots (11)$$

where the ζ, β refer to the like functions with the two sets of letters interchanged. Developing and reducing, this is

$$\phi(c+c') - \phi c - \phi c' = DF \text{ multiplied by}$$
$$3XX$$
$$+X(2avX + 5cv' + 8aX' - 15aa - 15a'a + 8)$$
$$+5v\,(\,vX - 3v' + a - 1)Y \cdot cz' - 3a' - 3a'X - 3a'X + 3aX' + 6aX'$$
$$-13(cc' + aa') + 9aav,$$

and thence

$$\phi = DF \text{ multiplied by}$$
$$X^4$$
$$+X(2cv - 10v' + 13c) - LX$$
$$-4cv + 25v' - M,$$

where the constants L, l, λ have to be determined. Does η being a cubic curve the number of triangles vanishes; that is, we have $\phi = 0$ for $a = 1, X = 0, \beta = 0$, and for $a = Z = 0, \beta = 0$ and $a = Z, \beta = 0$; we thus have

$$\begin{cases} a = 8, & X = 0,\ \beta = 10, \\ 0 = 88 - 4L - M - 15a, & \\ 0 = 88 - 4L - M - 15a, & \end{cases}$$

giving $L = 1, l = 16, \lambda = 3$. Whence finally,

$$\phi = (X^4 + X(2cv - 10v' + 13c - 1) \cdot c - Y + 2XZ - 16c - 3a^3]DF.$$

Second process; by reciprocation, we have

$$g - \chi - \chi' + F(c - c' - c') = 0,$$

$$c - c' = F(v - y - y') = 0.$$

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and thence

$$g - \chi - \chi' = DF(c + v - y' - y'),$$

Moreover

$$\chi = (X-2)e\,(X-1)F\,(c-1),$$

$$\chi = F\,(\,c-1)e\,(X-1)\,(X-1)\,a, \quad \chi = v\,a,$$

$$\chi + \chi' = DF\cdot(X-1)\,a\,(\,X-1)\,a,$$

and

$$v - y - y' = (X-X)\,a + 2(X-X)\,a - 3),$$

as is easily obtained, but see also post, No. 21; hence

$$g = DF \text{ multiplied by}$$

$$\{(X-1)+(X-1)\,(X-1)\}^2$$
$$+(X-1)\,(X-1)\,a^2;$$

but I reject the term $DF\,(\,X-1)\,a$ as not first giving a heterotypic solution; I do not go into the explanation of this. And then substituting for β its value, we have

$$g = DF \text{ multiplied by}$$

$$(X-1)\cdot\{a\,(\,v-1)\,(c-1)\}$$
$$+X^4 - X\,a + 5a - 3a^2,$$

where the second factor is

$$= X^4 + X(2cv - 10v' + 13c - 1)\cdot c - 4cv + 2c - 16c - 3a^3.$$

which is the foregoing result.

Case 40. $B = D = F = a = v$. By reciprocation of 39.

$$\chi = v\cdot a + v(X^4 + y(2Y - 10Y + 13Y - 1)\cdot Y - 4Y + 25Y - 16Y - 3a^3]F.$$

Case 41. $c = e = a$, $D = F = a$.

$$\chi = Bc(X-1)e\cdot D\,(X-1)F\,a, \quad \chi' = Fc(X-1)(c-1)Ba,$$

$$g = (X-1)cD\,(X-1)Ba + 2(X-1)\,(B-1)\,(X-1)\cdot ca\,(\,B-c-1)cBa,$$

= &c.

Case 42. $a = c = e = v$, $B = D = F = a = v$.

Functional Process; the curve a is supposed to be the aggregate of two curves, say $a = c + c'$, $B = D = F = v = v'$. Forming the construction…

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The enumeration is

Case		
$a\ Bc\ X\ d\ X$	$aBc\ X\ d\ X^3$,	(42)
$a'.a'Xa\,X$	ditto,	(11)
$a.a'Xa\,X$		(11)
$a.a'Xa\,X$		(12)
$a'Xc\,Xa\,X$		(17)
$a'Xc\,Xa\,X$		(10)
$a'Xc\,Xa\,X$		(10)
$a'Xc\,Xa\,X$		(10)

whence

$$\phi(c+c') - \phi c - \phi c' = \nu B \text{ multiplied by}$$

$$4(X-1)(Xa-X-a)\ldots\ldots(11)\times 2$$

$$+2\nu(a-1)X^2\ldots\ldots(12)\times 1$$

$$+4\nu(a-1)X(X-1)\ldots\ldots(17)\times 2$$

$$+5\nu(a-1)X^2\ldots\ldots(10)\times 2$$

$$+5\nu Xa(a-1)X + X(c-1) + 4\nu(Xa'+X'a) + 2\nu(X'a)X + \ldots\ldots(10)$$

where the ζ, β' refer to the like functions with the two sets of letters interchanged. Developing and reducing, we have

$$\phi(c+c') - \phi c - \phi c' = aB \text{ multiplied by}$$

$$X^4 \quad (8aa' - 5a^2a^2 - 6aa)$$

$$+XX'\{4aX(c-1) + 8a^2X + aa(a-2c+9) + (c-1)X\}$$

$$+X^3\{(2v' - 6a^2 + 15a^2)$$

$$+X^3\{12aa - 6aa' - 12a^2 + 18a\}$$

$$+X^2\{(-6a^2 - 12a + 15a^2)$$

$$+Saa^2,$$

and consequently

$$\phi = v \text{ multiplied by}$$

$$XY\,(4ac - 5a^2)$$

$$+X^4\{l\sigma a^2 + \nu c - 6a^2\}$$

$$+X^3\{(\beta - 5\nu) + \beta - Ja\},$$

where the constants L, l, λ have to be determined. The number of triangles vanishes when the curve α is a line or a conic, that is, $\phi = 0$ for $a = 1$, $X = 0$, $\beta = 0$, and for $a = 2, \beta = 2$; we thus have

$$0 = 83 - L - M,$$

$$0 = 80 + 2L + 12 - M,$$

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Moreover, the data being equicipanyl, the result must be so likewise; we must therefore have $L = l$. We thus obtain $L = l = u$, $M = 0$, $\lambda =$ but $aBL = u\bar{l}$, $u = Bp = u^2$.

Second process. by correspondence: form $a = c = e =$, $D = F = a$. We have also from the special consideration that the points B, F are given as the intersections of the curve a, by the joint of the joint c, which first pulse does not pass through a, we have

$$(I - \phi - \phi') + \nu(c - X - Y) = v\nu,$$

and by the consideration that c, D are given as intersections, c a double intersection, of the curve with the first point of the point c, which first pulse does not pass through a,

$$\nu - X - X + 2(v - c - y - y') = v\nu,$$

where

$$g - \chi - \chi' = \nu v - \nu B\nu,$$

and

$$v - c - y' - y' = \nu B\nu,$$

so that this is

$$g - \chi - \chi' = 4\nu B\nu,$$

$$= 4\nu(aB - X(X-1)) \cdot v.$$

Also

$$\chi = B\,(X-1)\,(X-1)\,(X-2)e\ \cdot Fa,$$

$$\chi' = F\,e\,D\,(X-1)\,X(X-1) \cdot a\,e\ \cdot \chi,$$

so that

$$g = B\nu \text{ multiplied by}$$

$$2(X-1) \cdot (X-1) \cdot (X-2) + 4\nu + 5z + R\,B;$$

viz. this is

$$B\nu(X^4 - X(X - X - X)\,(X-1)\,(X-1) + 8c) \cdot Xa + xa + e),$$

which is the number of positions of the angle (a), but several of these belong to special times of the triangle $aBcDeF$, giving heterotypic solutions, which are to be rejected; the required number is

$$[XY\,(X-1)\,(X-1)\,(X-1)\,(X-2)\,\nu^3 - \ a\,\nu^3\,] - \text{Red.}$$

Third process: form $c = a = B$, $a = Y\,e$.

$$g = g + \chi' - \text{Red.},$$

$$\chi = X - (X-1)(X-1)\,e\ \cdot v\nu,$$

$$\chi = X - (X-1)\,B\,Y\,(X-1) - Y\,cY,$$

$$\chi + \chi' = 2X^4\,(X-1)\,(X-1)\,Y\,(X-1) - Y\,cY.$$

The first-mode reduction is here

$$aB\,[X - (X-1)\,X - (X-1)4]\,.$$

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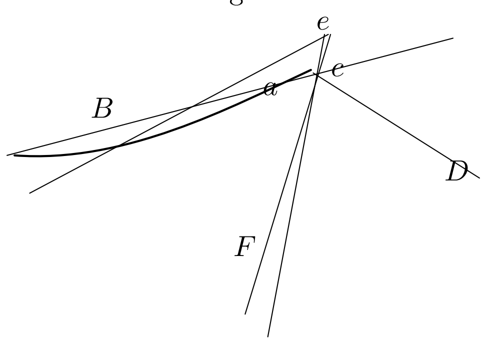
where the last term aBe arises from the tangents cBa and eFa, each coinciding with a cuspidal tangent, as shown in the figure.

Fig. 4.



The second-mode reduction is

$$aBX\,(X-1)\,(\,c-3a),$$

so that the two reductions together are

$$= aB\{(X-1)X+(X-1)(B-(X-1)k+e+e\,(\,c-2)(c-3a),$$

viz. this is

$$= aB\{X-1)X+4(B-1)+kX+e+a\{X-1)X+8c+5z+4a-3a-3a-2);$$

or substituting for Ω its value, and $-\Sigma X-3X+\delta\partial(\,e-\,)-2\partial(c\,-\,3a)$, and reducing, it is

$$aB\{X(X-1)(2a-c-4)-4ax+4a+12c-3a^2\}.$$

On account of the symmetry we must divide by $\frac{1}{2}g$.

Case 43. $a=c=e=D=F=a$.

Suppose for a moment that the angle a is a free point; the locus of a is a curve the order of which is obtained from Case 28, by writing $c=e=v=v$, $B=D=F=u$; the locus in question meets a curve a in $u(2X^2+2X-1)\,e\cdot a$ points, which is the intersection of the curve. Suppose that a is a given point, and several of these belong to special times of the triangle $aBcDeF$, giving heterotypic solutions, which are to be rejected; the required number is

$$[2X(X-1)(X-1)(X-2)\,\nu^3-\,a\,\nu^3\,]-\text{Red}.$$